Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make each partial a separate controlled experiment.

Show every parameter vector and full denominator;document numerical assumptions and keep directional changes separate from partials.

Keep one habit: Assess independent loss probes,formula agreement and coordinate order with checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Estimate one **partial derivative** while other parameters stay fixed

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The two evaluation points lie one step on either side of the centre, so their separation is two steps.

Opening retrieval

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate an average rate of change over a stated interval

Reference example, not the only acceptable wording: Teacher explanation: Average slope is (0 - 4) / 2 = -2. The local slope at 2 is zero. They describe different locations or intervals.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish architectural choices from parameter values

Reference example, not the only acceptable wording: Teacher explanation: There are 8 weights and 2 biases, total 10.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Look for: Trace identity ReLU and sigmoid on a numerical score

Reference example, not the only acceptable wording: Teacher explanation: Identity gives -2, 0, 2; ReLU gives 0, 0, 2. ReLU retains the positive score itself; it is not a step classifier.

Pictured end goal

By the end, I can explain **gradient**s with a numerical check using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Slopes and local change

Parameters and architecture

Activation functions

For the compact alternative schedule, use two 45-minute sessions to separate coordinate probes from combined changes. The first **gradient** check is deliberately an affine model with squared loss, avoiding hidden-layer calculus.

Introduce the chain-rule reasoning in words before showing the compact formula.

Outcomes

Estimate one **partial derivative** while other parameters stay fixed

Assemble a **gradient** in a declared parameter order

Compare numerical and formula based **gradient**s with a **tolerance**

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach one-coordinate probes before assembling the gradient

> Make the full parameter vector visible at every loss evaluation.

Main model,complete perturbations and guided key

Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

There are two parameters in order [w, b].

To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

The helper assumes finite numerical **input**s and a positive nonzero step. The formula follows the squared-loss rule: local loss change is twice the residual times the **prediction** sensitivity.

Changing w by a small amount changes **prediction** twice as much here; changing b changes **prediction** by the same amount.

Exact floating-point equality is not required. Record the step and a **tolerance**, then try more than one parameter setting.

A passing numerical check supports agreement for those tested points; it does not prove all possible **input**s.

Main probe from centre[w, b]=[1, 0] · Minus parameter vector /prediction /loss · Plus parameter vector /prediction /loss · Full gap in probe variable · Slope

> **Main probe from centre[w, b]=[1, 0]:** w only; hold b0
>
> **Minus parameter vector /prediction /loss:** [.9, 0]/1.8/1.44
>
> **Plus parameter vector /prediction /loss:** [1.1, 0]/2.2/.64
>
> **Full gap in probe variable:** 0.2
>
> **Slope:** -4

> **Main probe from centre[w, b]=[1, 0]:** b only; hold w1
>
> **Minus parameter vector /prediction /loss:** [1, –.1]/1.9/1.21
>
> **Plus parameter vector /prediction /loss:** [1, .1]/2.1/.81

> **Full gap in probe variable:** 0.2
>
> **Slope:** -2

> **Main probe from centre[w, b]=[1, 0]:** Both via t:[w, b]=[1+t, t]
>
> **Minus parameter vector /prediction /loss:** [.9, −.1]/1.7/1.69
>
> **Plus parameter vector /prediction /loss:** [1.1, .1]/2.3/.49
>
> **Full gap in probe variable:** 0.2
>
> **Slope:** -6

For this one-example squared loss, residual e=2w+b−3. The formula gradient is[2e×2, 2e]=[4e, 2e]in parameter order[w, b]. At the original[1, 0], e−1 gives[−4, −2]. Each numerical entry changes one coordinate while the other stays at its original value. The combined probe varies t along[1+t, t]; its slope−6is with respect to t, not either partial or slope per unit Euclidean distance.

The checker uses spacing.001 and absolute tolerance 1e−6; it assumes valid finite numbers and positive spacing rather than validating them.

Guided: **prediction** 1, residual -2, loss 4, **gradient** [-8, -4]. Exit: changing both parameters measures a directional change; each partial must vary just its own coordinate while holding the others fixed.

Complete pictured same-model facts and scope

Check each coordinate of a **gradient** while holding the other parameter fixed.

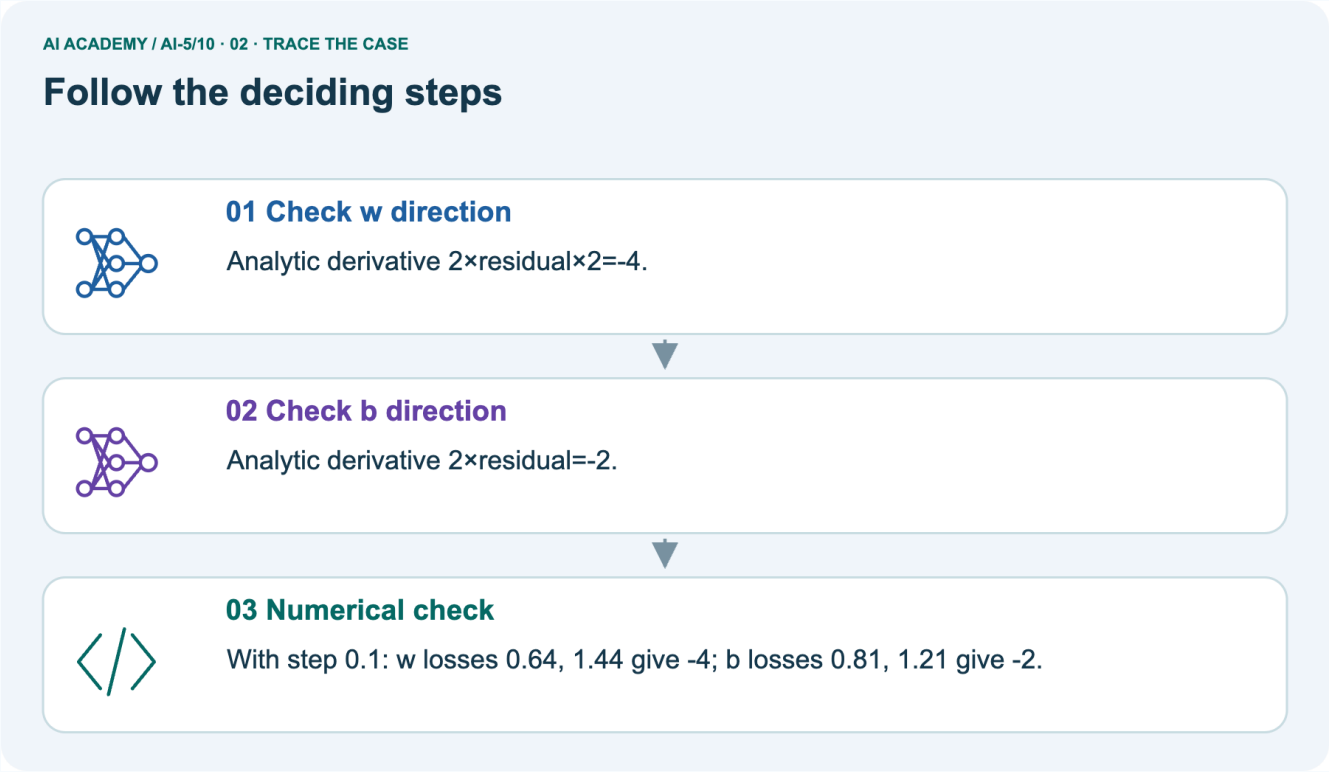
Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses the same model y=2w+b, target 3, centre[1, 0], residual−1. With step.1, w-only minus/plus losses 1.44/.64 give−4; b-only losses 1.21/.81 give−2. Order[w, b]therefore gives[−4, −2]. Returning[−2, −4]mislabels both entries even though the numbers look familiar.

The fixed x2 makes w twice as sensitive as b here; the independent exercise supplies x3 and a different target, so its formula must use that new input.

Trace labelled partials and component-swap failure



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check w direction	Analytic derivative 2×residual×2=-4.
2. Check b direction	Analytic derivative 2×residual=-2.
3. Numerical check	With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

Change: Swap the two **gradient** components in the program **output**.

Prediction: The coordinate-wise numerical check exposes the mismatch.

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Partial derivative	A local slope with respect to one parameter while others are held fixed
Gradient	The ordered vector of partial derivatives of a scalar function
Finite difference	An estimate of change computed from nearby function values
Tolerance	A stated allowance for numerical disagreement

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

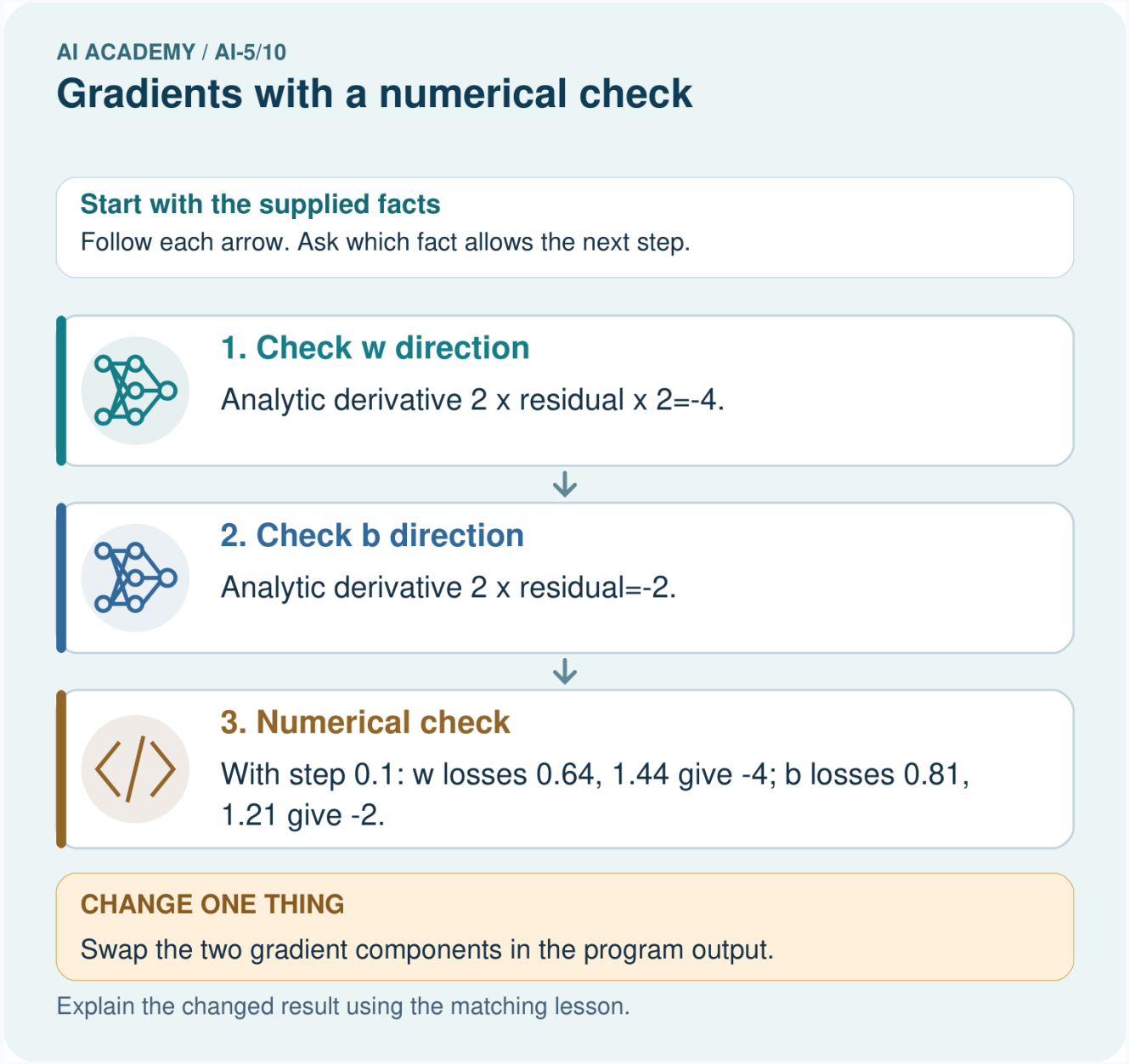
At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Gradients with a numerical check

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

Teacher check: Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: It has one entry per parameter for a scalar loss, in a stated order.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

They estimate a **partial derivative**. Changing both coordinates would mix their effects and would not independently verify [-4,-2]. Changed-case check: The coordinate-wise numerical check exposes the mismatch.

The w perturbation changes y twice as strongly as the same b perturbation.

Changed-case reasoning: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate w-probe and b-probe tables. Extension: at the lesson model [1, 0], test steps 0.1, 0.01 and 0.001 and report absolute disagreements from [-4, -2], without claiming universal **tolerance** guarantees.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the symmetric denominator and compute the initial prediction and residual.

Time or stage	Teacher action
7 to 18 minutes	Probe w alone with b fixed; record the two losses and slope.
18 to 32 minutes	Probe b alone and assemble [-4, -2]. Complete W1 and W2.
32 to 45 minutes	Trace the optional numerical checker and the guided [0, 1] case.
Session 2 start to 8 minutes	Use W3 to distinguish a combined direction from two partial derivatives.
8 to 20 minutes	Discuss the ReLU corner in W5 and a stated numerical tolerance.
20 to 35 minutes	Collect W4 independently, requiring both pairs of losses and coordinate order.
35 to 45 minutes	Use the exit response and fresh retry, then connect local gradient evidence to next week.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main coordinate checks and contracts as supplied. W2 fullgap.2 gives−4. W3 combined-direction slope −6 with respect to t, not either partial or normalized-distance slope. W5ReLUcornerdifference.5 doesnotestablish derivative. Independentw0, b1:y1, e−1, L1; wprobesy.7/1.3 loss1.69/.49, partial−6; bprobesy.9/1.1 loss1.21/.81, partial−2; formula[2e×3, 2e]=[−6, −2]. Keep coordinate order and numerical tolerance.

Diagnose the current misconceptions

The **gradient** is one scalar: It has one entry per parameter for a scalar loss, in a stated order.

Use the updated weight for the bias probe: Both entries must refer to the same current parameter vector.

A symmetric value defines the derivative at any corner: Left and right slopes can differ; the ReLU example exposes this.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

Teacher explanation: Residual -1 and loss 1. w probes give **prediction**s 0.7 and 1.3, losses 1.69 and 0.49, partial -6. b probes give **prediction**s 0.9 and 1.1, losses 1.21 and 0.81, partial -2. Formula **gradient** is [-6, -2].

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Recover a gradient

For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

Teacher explanation: **Prediction** 3, residual 0, loss 0, **gradient** [0, 0].

W2 Catch a denominator error

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

W3 Separate a direction from a coordinate

Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

Teacher explanation: Residuals are -1.3 and -0.7; losses 1.69 and 0.49. Slope is -6. It measures the combined direction [1, 1], equal here to the sum of partials -4 and -2.

W5 Recognise a corner

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

Teacher explanation: The value is 0.5. The left slope is 0 and right slope is 1, so the slopes disagree and the derivative at zero does not exist.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: **prediction** 1, residual -2, loss 4, **gradient** [-8, -4]. Exit: changing both parameters measures a directional change; each partial must vary just its own coordinate while holding the others fixed.

W1 Recover a gradient

W2 Catch a denominator error

W3 Separate a direction from a coordinate

W4 Check a new example independently

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

Teacher explanation: Residual -1 and loss 1. w probes give **prediction**s 0.7 and 1.3, losses 1.69 and 0.49, partial -6. b probes give **prediction**s 0.9 and 1.1, losses 1.21 and 0.81, partial -2. Formula **gradient** is [-6, -2].

W5 Recognise a corner

Independent assessment and fresh retry

W4 requires numerical and formula **gradient**s [-6, -2] with losses shown. Fresh retry: x = 1, target 4, w = 2, b = 1 gives residual -1 and **gradient** [-2, -2]. Step 0.1 in either coordinate gives losses 1.21 and 0.81.

Require the fixed coordinate to be named.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate w-probe and b-probe tables. Extension: at the lesson model [1, 0], test steps 0.1, 0.01 and 0.001 and report absolute disagreements from [-4, -2], without claiming universal **tolerance** guarantees.

Week 11 applies **gradient**s to simultaneous parameter updates.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/11, ai-5/13, ai-5/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Estimate one **partial derivative** while other parameters stay fixed

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

Reasoned practice answer: Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

Check every learner before discussion: Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: It has one entry per parameter for a scalar loss, in a stated order.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Swap the two **gradient** components in the program **output**. Require a **prediction** and a reason before comparing. Expected result: The coordinate-wise numerical check exposes the mismatch.

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

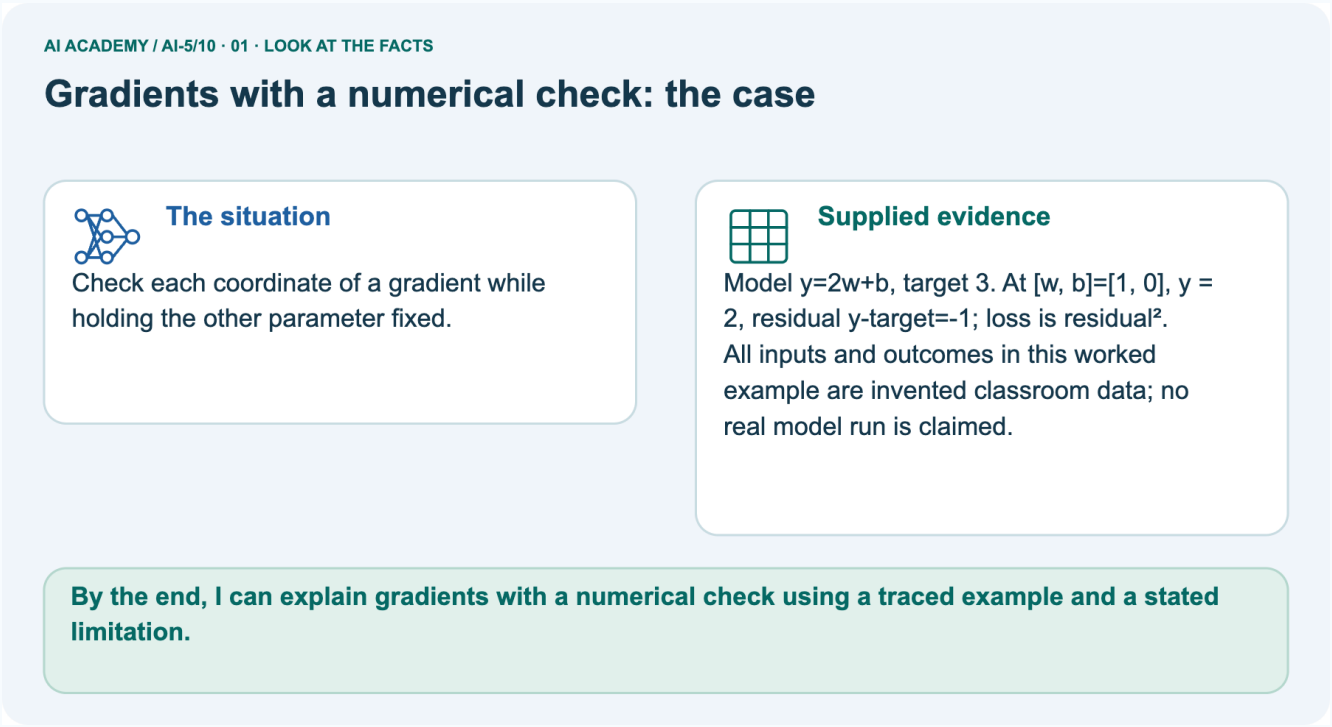
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Gradients with a numerical check: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

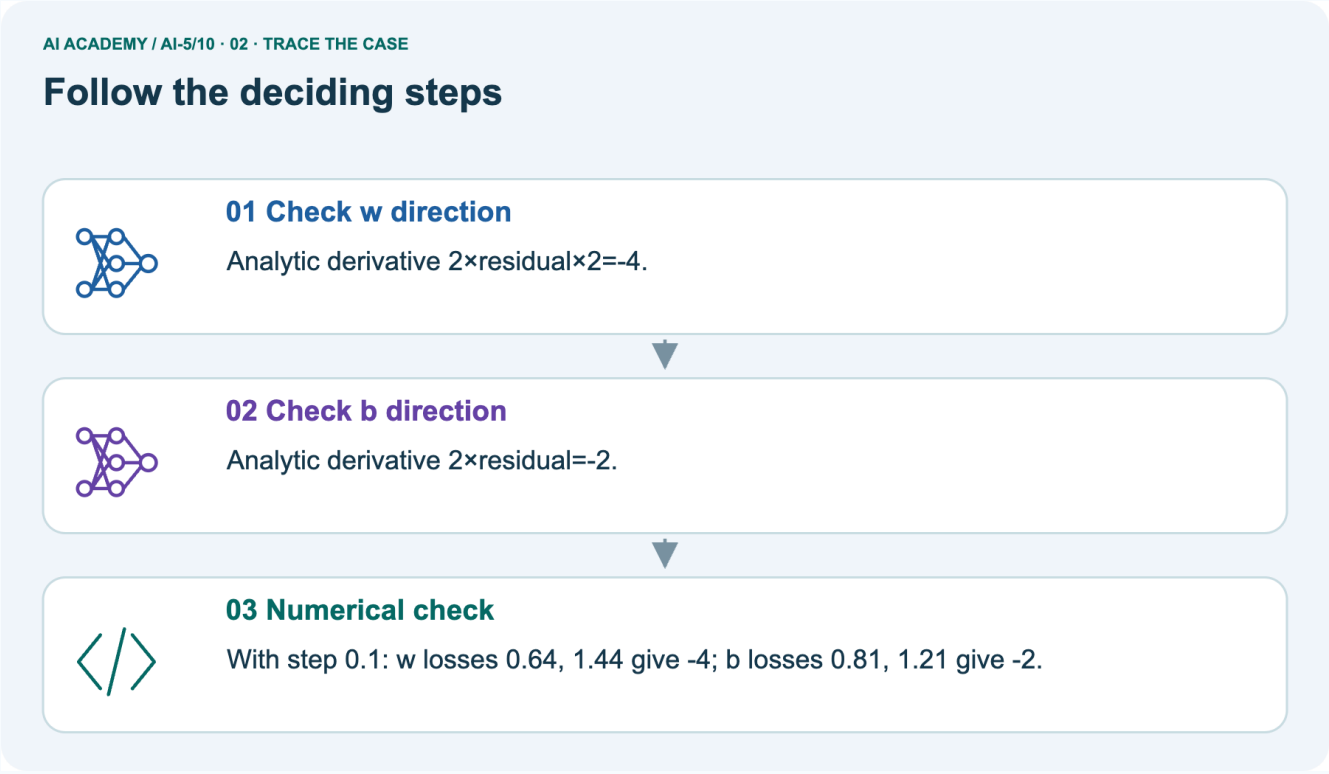
Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



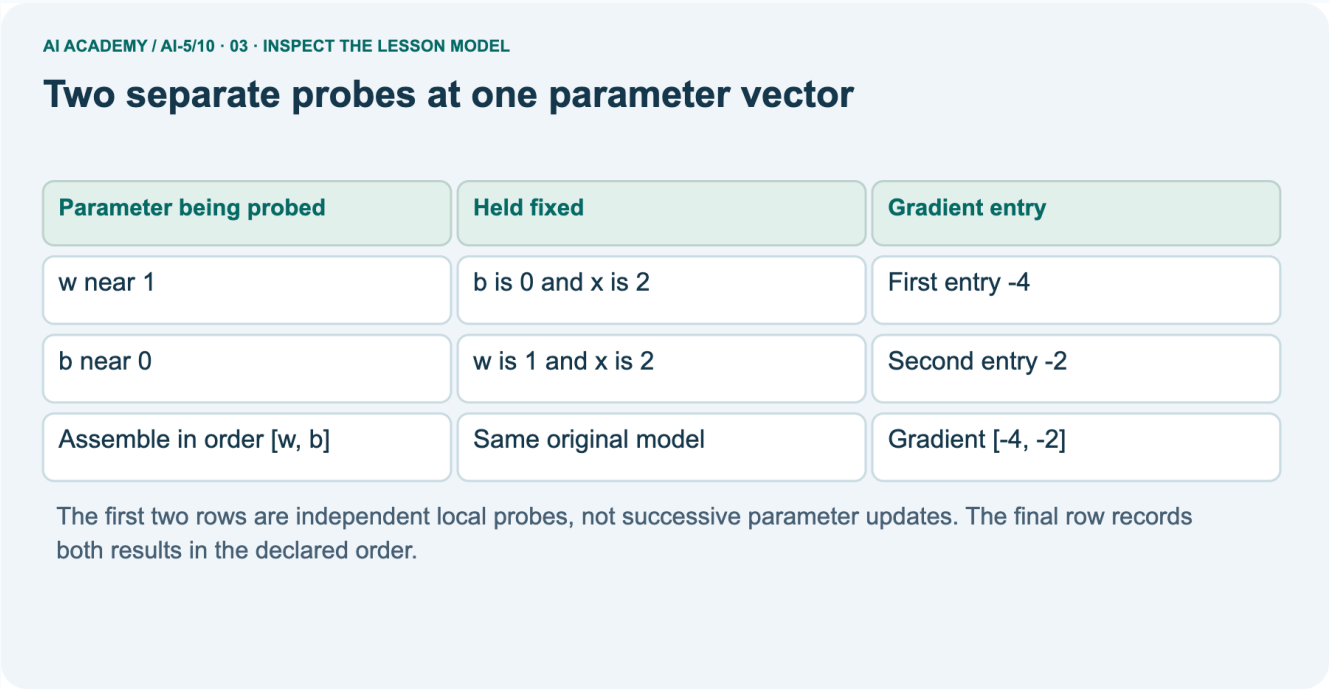
Follow the labelled picture with the matching explanation. Open the vector diagram

Check w direction: Analytic derivative 2×residual×2=-4. → Check b direction: Analytic derivative 2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two separate probes at one parameter vector



AI ACADEMY / AI-5/10 · 03 · INSPECT THE LESSON MODEL

Two separate probes at one parameter vector

Parameter being probed	Held fixed	Gradient entry
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

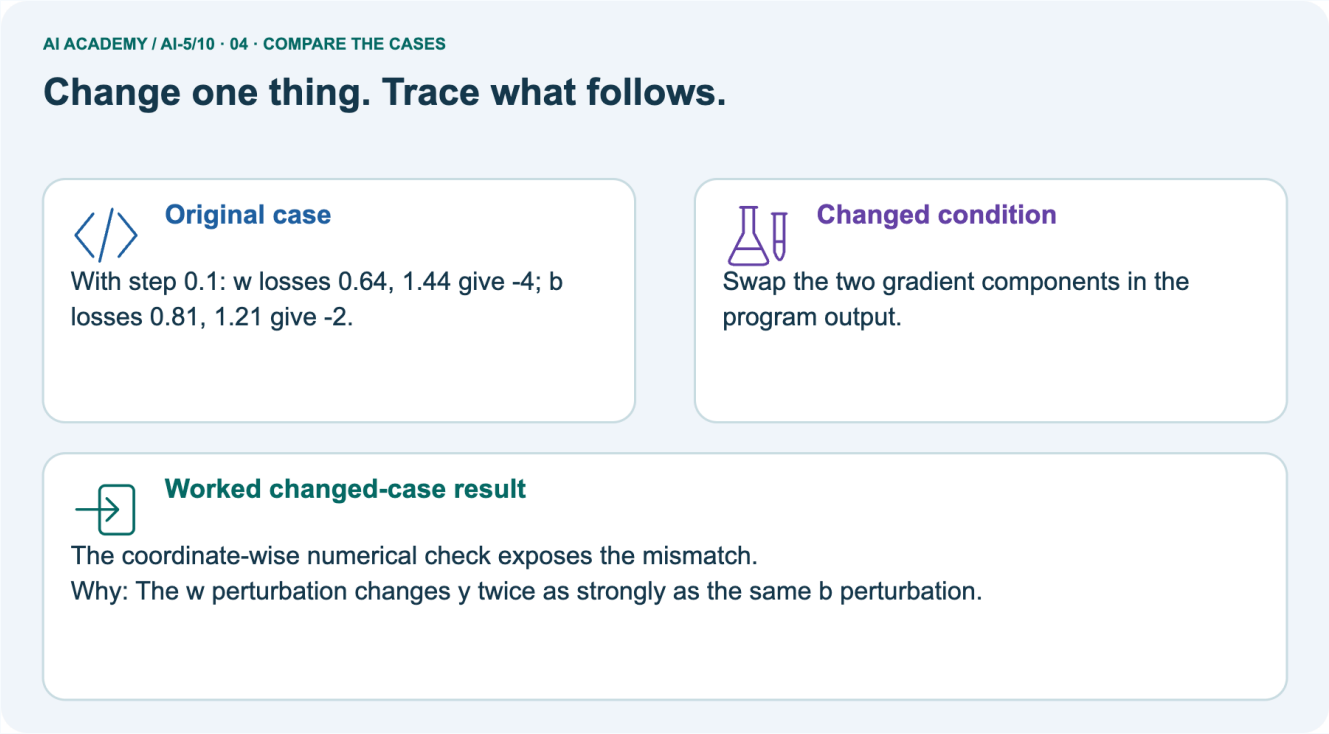
Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

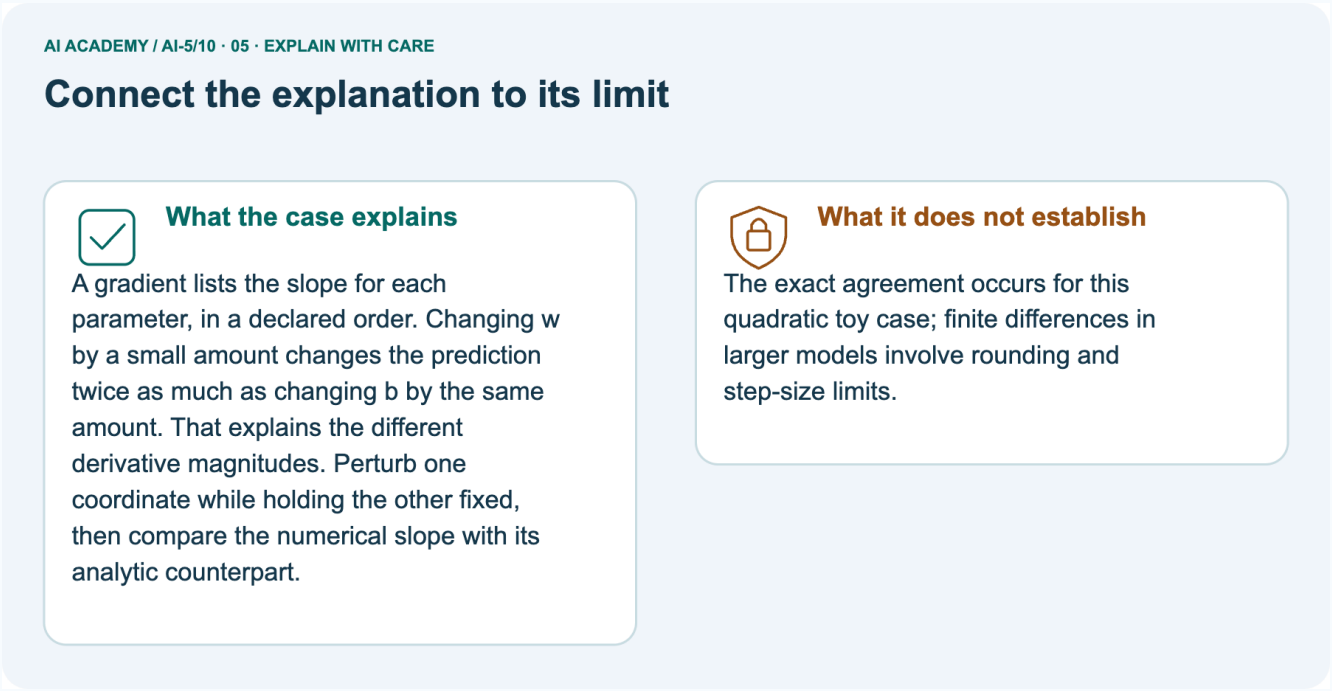
Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The w perturbation changes y twice as strongly as the same b perturbation.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

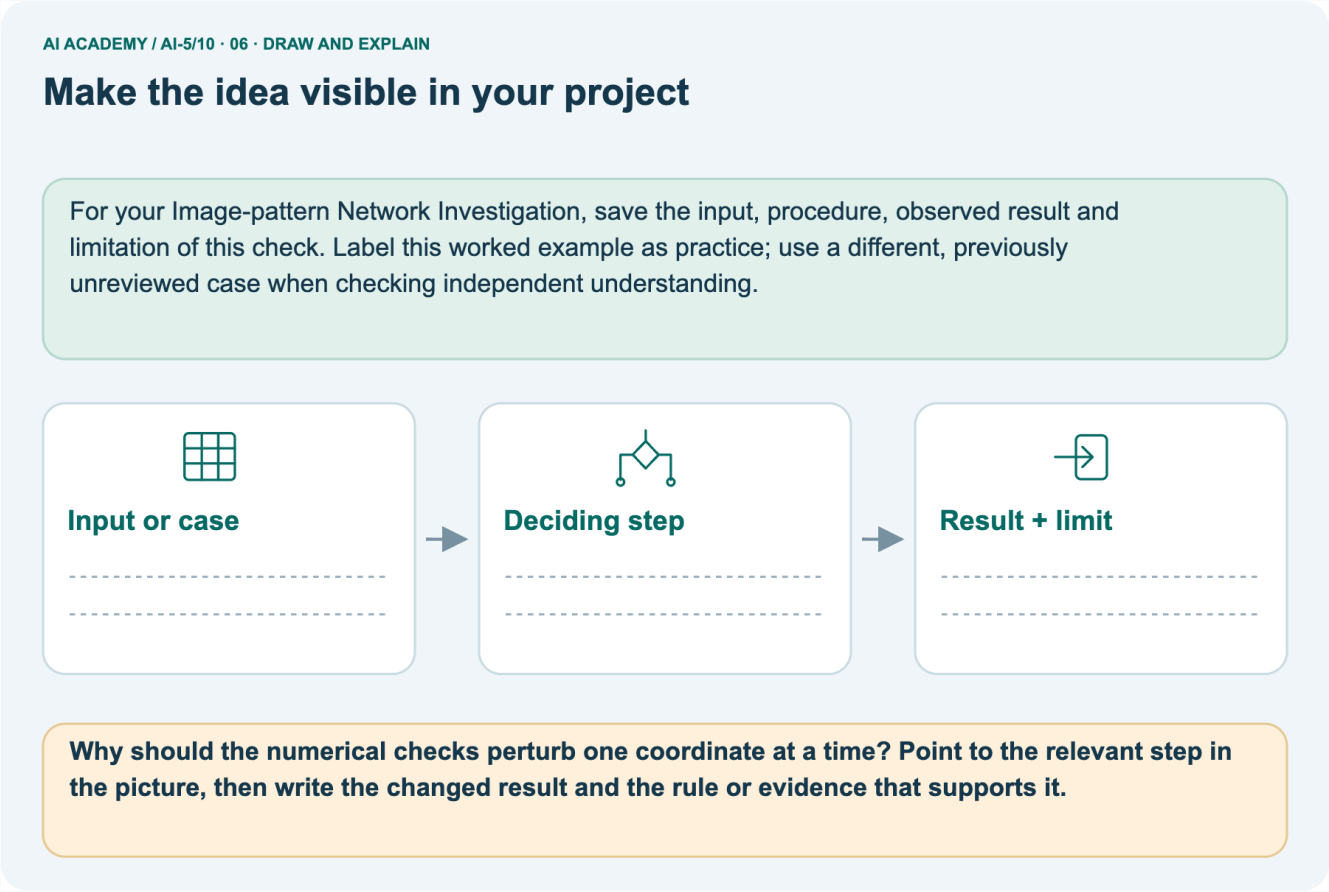
Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: They estimate a **partial derivative**. Changing both coordinates would mix their effects and would not independently verify [-4,-2].

Changed-case check: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Gradients with a numerical check: the case

02. Follow the deciding steps

03. Two separate probes at one parameter vector

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.